20.11.2023

EE2003 and CSE 3017 DIGITAL DESIGN MIDTERM EXAM

Name:

CLOSED BOOK, CLOSED NOTES EXAM. NO CALCULATORS OR CHEAT SHEETS ALLOWED.
PLEASE PROVIDE YOUR ANSWERS IN DESIGNATED PLACES.

Q1) (20 points) A combinational circuit is represented by the truth table below. Describe this circuit in Verilog using dataflow modeling. The first output v will be represented in SOP form. The second output w will be represented in POS form. Minimization is not required.

Inputs			Outputs	
x	y	z	v	w
0	0	0	1	0
0	0	1	0	0
0	1	0	0	1
0	1	1	0	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	0
1	1	1	1	0

Q2) (10 points) When the below binary subtraction operations are done in Verilog under Vivado, what will be the result?

	0	1	0	1
-	1	0	1	0

	1	0	1	0
-	0	1	0	1

Q3) (10 points) When the below Verilog description and its associated testbench file is executed in Vivado, what will be the output in hexadecimal form?

```
module test(num1,num2,num3,res1);

input [11:0] num1, num2, num3;
output [11:0] res1;

assign res1[11:8]=num1[11:8];
assign res1[7:4] =num2[7:4];
assign res1[3:0] =num3[3:0];

endmodule
```

```
`timescale 1ns / 1ps
module test_tb;
reg [11:0] in1, in2, in3;
wire [11:0] out1;
test UUT(.num1(in1),.num2(in2),.num3(in3),.res1(out1));
initial begin
in1 = 12'hABC;
in2 = 12'b101011110000;
in3 = 12'd2730;
#100;
end
endmodule
```

out1 in hexadecimal form

Q4) (20 points) You are working with floating-point numbers in your digital system and asked to design a format converter from “half” to “single” form. Implement a Verilog module for the conversion operation. Your module gets input X1 in SEF form and gives output X2 again in SEF form.

Q5a) Design a negative/positive number detector with the following specifications. Input to the system is an eight-bit number. If the input number is negative, assume it is given in two’s complement form. If the number is positive, the first output will be logic level 1. Otherwise, the second output will be logic level 1.

A) **(10 points)** Implement the designed combinational circuit in Verilog.

B) **(10 points)** Sketch the logic circuit.

Q5b) Design a magnitude displayer with the following specifications. It will be single input single output circuit. Input to the system is an eight-bit number which can be positive or negative in two’s complement form. Output will display the magnitude of the input number. Hint: If the input number is negative, you can find its magnitude by converting it back from two’s complement form.

A) **(10 points)** Implement the designed combinational circuit in Verilog.

B) **(10 points)** Sketch the logic circuit.